# Multimedia Signal Processing - 2nd Module

February, 18 2021

1. **Question (8 pts)** Consider a continuous source $X$ that emits the following series of symbols

$$x = [\sqrt{2},\ \sqrt{2},\ -2\sqrt{2},\ -2\sqrt{2},\ -\sqrt{2},\ -\sqrt{2},\ 2\sqrt{2},\ 2\sqrt{2}].$$

You are given a quantizer $Q(\cdot)$ that applies the following quantization rule

$$Q(x) = \left\lfloor \frac{x}{\Delta} \right\rceil \cdot \Delta,$$

where $\lfloor \cdot \rceil$ rounds to the closest integer, and $\Delta = 3$ is the quantization step. Additionally, you are given the transform matrix

$$T = \begin{bmatrix} \sqrt{2}/2, & \sqrt{2}/2 \\ \sqrt{2}/2, & -\sqrt{2}/2 \end{bmatrix}$$

   (a) Compute the probability mass function of $x$ directly from the sequence.

   (b) Compute the entropy of $x$.

   (c) Compute a possible Huffman code to encode $x$

   (d) Compute the average code length of the Huffman code, and compare it with the Shannon code length.

   (e) Check that the matrix $T$ is orthogonal, thus can be used for transform coding.

   (f) Consider a transform coding scheme that: (i) select a block of two samples; (ii) apply the matrix $T$; (iii) quantize the obtained coefficients with the quantizer $Q$.

      i. Compute the sequence $y$ of transform coefficients obtained applying $T$ to $x$ (hint: consider non-overlapping blocks).

      ii. Compute the sequence $\tilde{y}$ obtained quantizing $y$ with $Q$.

2. **Question (7 pts)**
   Oral examination.

3. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Consider a source $X$ with alphabet $A = \{a, b, c, d\}$. Consider the following associated code: $C(a) = 0$, $C(b) = 10$, $C(c) = 110$, $C(d) = 111$. This code may have been obtained using Huffman coding scheme.
   - [T] [F] Mutual information between two discrete sources $X$ and $Y$ can be computed starting from joint entropy $H(X, Y)$ and conditional entropies $H(X|Y)$ and $H(Y|X)$.
   - [T] [F] Compander-expander quantization method is better than a uniform scalar quantizer to quantize uniformly distributed signals.
   - [T] [F] The optimal bit allocation strategy for transform coding may lead to results that are not practically feasible.
   - [T] [F] Consider a video coding scheme based on motion compensation. The decoder must know the algorithm used by the encoder to estimate motion vectors.
   - [T] [F] Consider an image coding scheme based on 2D-DCT block transform coding. The DC component of each block is proportional to the average pixel value of the block.
   - [T] [F] Given a discrete source $X$ characterized by entropy of 3 bit/symbol, it is possible to find a code with at least one codeword shorter than 3 bits.
   - [T] [F] In a speech coding scheme, the long-term predictor should be tuned according to the pitch of the voice.

4. **Question (Matlab - 10pts)** Consider the case of computing LPC synthesis of a **voiced** audio segment. To do this, download the audio segment 'voiced.wav' to the current folder by running the command `websave('voiced.wav', 'https://bit.ly/2NCugn4');` and follow these steps:
   (a) Open the audio file, consider only the first 200ms frame.
   (b) Plot the absolute value of the DFT [dB] of the signal as a function of the frequency [Hz], starting from frequency 0 [Hz].
   (c) Determine the pitch [Hz] of the frame, considering only frequencies between 60 Hz and 500 Hz.
   (d) Compute LPC coefficients of order $P \in \{2, 4, 8, 16, 32, 64, 128\}$.
   (e) For each $P$, compute the prediction error using the corresponding coefficients.
   (f) For each $P$, compute the LPC synthesis of the frame. Normalize the range of the synthesized frame in $[-1, 1]$. Explain how did you chose the input signal to the filter, considering that you do not use the prediction error.
   (g) For each $P$, compute the SNR considering the original frame as signal, and the MSE between the original and the synthesized frame as noise.
   (h) Plot the rate distortion curve, using $P$ as rate and SNR as distortion. Can you draw some conclusions? Why?